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RESEARCH ARTICLE

Phosphorus Sources Combined with Doses of Organic Compost Increased the Population of Soil Microorganisms and P Level in the Soil and Plant and the Dry Matter of Sugarcane

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Abstract The aim of the study was to evaluate the effect of P sources and filter cake, enriched or not with biofertilizers and liquid peat, on the microbial population and the soil available P levels and shoot dry matter and P accumulation by sugarcane. The experiment was conducted in a greenhouse in a completely randomized design with factorial arrangement $4 \times 3 + 1$ and three replications. The treatments consisted in four sources of P: simple superphosphate, triple superphosphate, monoammonium phosphate and reactive natural phosphate combined with three levels of organic compound: without filter cake, decomposed filter cake (30 t ha^{-1}, dry basis) and decomposed filter cake enriched with liquid peat (4 mL kg^{-1} of filter cake) and biofertilizers (480 mL kg^{-1} of filter cake) and a control treatment without organic compound and without phosphorus. At 45 and 110 days after planting the total population of bacteria and fungi, phosphate-solubilizing microorganisms and total microorganisms were evaluated. After 60 and 110 days, the soil available P, plants' P and dry matter and P accumulation were evaluated. The organic compound increased the population of soil microorganism, the soil P levels, the accumulation of P and dry matter by the plants, regardless of the source of P, although the enrichment with biofertilizer and liquid peat did not enhance this effect. In conclusion, the presence of organic compost enhances the effects of phosphate fertilization.

Keywords Microorganisms · Organic compound · *Saccharum* spp. · Soil biology

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Introduction

Phosphorus (P) deficiency is the second nutritional disorder that causes worldwide restriction on plant growth (Khan and Joergensen 2009), with greater importance in tropical soils due to the low availability and processes of adsorption and precipitation. P is the second nutrient that most limits crop productivity (Biswas and Narayanasamy 2006); in cases of P deficit, its availability to crops can be increased by the use of organic or inorganic fertilizers (Ayaga et al. 2006). However, 85 to 90% of the added inorganic P becomes unavailable to the plant in the application year due to adsorption and precipitation with Fe, Al and Ca in the soil (Gichangi et al. 2009).

Thus, the increase in P fertilization efficiency has been sought. Studies demonstrate that the combination of a phosphate fertilizer source with organic compounds shows an increase in the availability of this nutrient in the soil (Gichangi et al. 2009; Santos et al. 2009; Takeda et al. 2009; Krey et al. 2013) and increase in soil microbial growth and activity (Nahas et al. 1994; Jilani et al. 2007; Stark et al. 2007; Jiang et al. 2011; Zhang et al. 2012; Krey et al. 2013), resulting in enhanced nutrient uptake and

productivity by the plant (Ezz et al. 2011; Almeida Júnior et al. 2011).

Many recent studies are pointing biofertilizers as another alternative. Biofertilizers are supplementary component to soil and crop management, whose application can significantly be useful in maintaining the sustainability of various crop productions. The P-solubilizing microorganisms are some of the plant-growth-promoting rhizobacteria that can be classified as biofertilizers (Sahoo et al. 2013). Other authors reported that this technology is cheap and provides increased levels of P in soil (Biswas and Narayanasamy 2006; Chen et al. 2006; Abou-el-Seoud and Abdel-Megeed 2012), higher nutrient plant uptake and productivity (Stamford et al. 2006; Singh and Reddy 2011).

Sugarcane filter cake (residue of the sugarcane industry) has the potential to partially replace mineral fertilizer in crops (Santos et al. 2010; Almeida Júnior et al. 2011; Yang et al. 2013). Although there are few studies that show the effect of organic and inorganic fertilizers and their combinations on the changes in soil microbial communities of sugarcane crops, there are reports in soils cultivated with rice (Zeng et al. 2007; Zhang et al. 2012). Studies suggest that the microbial community structure changes depending on the type of fertilization (Zhang et al. 2012), and the application of organic fertilizer can rapidly increase the total quantity of viable microorganisms and thus improve the ecological environment of the soil (Stark et al. 2007).

The primary mechanisms attributed to phosphate solubilization are the excretion of H^+ and organic acids. The phosphate-solubilizing microorganisms are known to produce organic acids such as citric acid, oxalic, tartaric, acetic, lactic, gluconic, among others (Babana and Antoun 2006) which are capable to dissolve mineral phosphate and make it available for plants. Besides, through the carboxyl and hydroxyl groups, organic acids are capable to chelate ions complexed with phosphate, such as Ca^{2+}, Fe^{2+}, Fe^{3+} and Al^{3+} (Vassilev et al. 2006).

Therefore, changes in the availability of soil P are governed by chemical and biological processes and are still influenced by the fertilizer sources. Thus, it is proposed to evaluate the effects of addition of different organic compounds to inorganic sources of phosphate fertilizers compared with only the use of fertilizer. In addition, it is suggested that this effect can be enhanced by supplementing the organic compound and the fertilizer with biofertilizer and peat.

In this scenario, it is pertinent to test the hypothesis that the effects of phosphate fertilization can be decisively influenced by the source and associated materials such as organic compounds, biofertilizers based on microorganisms and peat. The aim of the study was to evaluate the effect of P sources and organic compound (filter cake) enriched or not with biofertilizers and liquid peat on the microbial population and the soil available P levels and dry matter and P accumulation by sugarcane plants.

Materials and Methods

The experiments were carried out at Faculdade de Ciências Agrárias e Veterinárias/UNESP, Jaboticabal, Sao Paulo, Brazil, from March to June of 2012, in greenhouse conditions, with RB85-5156 sugarcane variety.

The soil used to fill the vessels was the clayey Oxisol (Embrapa 2013) collected from a layer from 0.20 to 0.50 m, which showed the following chemical characteristics (Raij et al. 2001).We used the soil collected from a layer from 0.20 to 0.50 m with the purpose to decrease the influence of anthropic activities in the soil, since it was agricultural soil. The chemical characteristics of the soil were determined as described by Raij et al. (2001) (Table 1).

The experiment was conducted in a greenhouse in a completely randomized design with factorial arrangement $4 \times 3 + 1$ and three replications. The treatments consisted in four sources of P: simple superphosphate, triple superphosphate, monoammonium phosphate and reactive natural phosphate combined with three levels of organic compound: without filter cake, decomposed filter cake (30 t ha^{-1}, dry basis) and decomposed filter cake enriched with liquid peat (4 mL kg^{-1} of filter cake) and biofertilizers (480 mL kg^{-1} of filter cake) and a control treatment without organic compound and without P. The dose of P applied was 25 mg dm^3 of P soluble in 2% citric acid. The experimental unit consisted of a vessel with capacity for 20 dm^3 (diameter 40 cm; height 30 cm) with two plants.

The enriched organic compound was obtained by composting filter cake, with the addition of biofertilizer containing natural compounds, polysaccharides, organic acids of low molecular weight, humates, purified phosphate, apatite and selected microorganisms with a load of 1×10^6 colony-forming units (CFU) per cm^{-3} of the product and liquid peat containing humic acid. During the experiment setup, the chemical analysis of filter cake was performed and the results were (values expressed in dry matter at 60–65 °C): pH ($CaCl_2$ 1 mol L^{-1}) = 8.2; total organic matter = 29.6%; total N = 1.4%; total P = 1.2%; total K = 0.2%; total Ca = 2.7%; total Mg = 1.1%; total S = 0.2%; and C/N ratio = 12/1.

In all treatments, nitrogen (200 mg dm^{-3}) in the form of ammonium sulfate (20% of N) and urea (45% of N) and potassium (150 mg dm^{-3}) in the form of potassium chloride (60% of K_2O) were uniformly applied. It should be emphasized that treatments with the absence of filter cake were balanced with N and K according to the amount that would be supplied by the filter cake, since these are the

Table 1 Soil characteristics from the layer of 0.20 to 0.5 m

pH $CaCl_2$	Organic matter g dm^{-3}	P (resin) mg dm^{-3}	K $mmol_c$ dm^{-3}	Ca	Mg	H + Al	Sum of bases	Cation exchange capacity	Base saturation %
5.5	11	8	1.1	20	9	18	30.1	48.1	63

nutrients that could potentially promote greater indirect effects. Ca and Mg levels were high during the entire experiment.

After the vessels were filled with the appropriate treatments, they were left for five days receiving periodic irrigation and then two plants of sugarcane were transplanted per vessel. To obtain seedlings, buds of approximately 4 cm were cut and then placed in polypropylene glasses (700 mL) containing washed sand to sprout and grow until they reached about 15 cm height. Daily irrigation was performed to maintain the humidity of 60% to 70% of the water-holding capacity.

At 45 and 110 days after the experiment started, samples of each vessel were collected to quantify the population of total bacteria and fungi, phosphate-solubilizing microorganisms and total microorganisms. To count the microorganisms, the serial dilution procedure (Wollum 1982) was used, using culture medium (Bunt and Rovira 1955) at pH 7.4 for total bacteria counting and culture medium (Martin 1950), pH 5.6 with penicillin 60 mg mL^{-1}, streptomycin 40 mg mL^{-1} and rose Bengal 70 µg mL^{-1}, for the total fungi counting. To quantify the total microorganisms and the phosphate-solubilizing microorganisms, the Pikovskaya culture medium (Martínez et al. 2006) was used. In addition, in this culture medium, the total microorganisms and the phosphate-solubilizing microorganisms counting was performed. The plates were incubated at 30 °C for 24 h (bacteria), 72 h (fungi) and 96 h (P-solubilizing microorganisms and total microorganisms) and then were evaluated on a CFU counter with 6 × magnification. To count the microorganisms, we did the dilution of 10^{-5} for bacteria, 10^{-3} for total fungi and 10^{-3} for solubilizing microorganisms.

At 60 and 110 days, the levels of available P in the soil were determined (Raij et al. 2001). At the same period, the shoots of each vessel's plants were collected. The collected plant material was washed and placed in paper bags and sent to the forced air circulation furnace at 60 to 65 °C until it reaches constant mass. After drying, samples were weighed and the dry mass of shoots per vessel was determined. Subsequently, the samples were ground in a Willey mill and the chemical analysis of P in the aerial parts of the plants was performed (Bataglia et al. 1978). The accumulation of P per vessel was calculated based on the dry matter production of each vessel and its P content.

The obtained data were subjected to homogeneity and normality tests, and, in the necessary cases, the data were converted to logarithm. The F test at 5% of significance was used to compare treatments, and the means were compared by the Scott-Knott test ($P < 0.05$). All the statistical tests were realized using the statistical program Assistant.

Results

The association of P sources with the organic compound showed no significant interaction ($P > 0.05$) at 45 days after the experiment installation (Table 2). The P sources promote alterations on the total bacterial population ($P < 0.01$), but the P sources did not promote alterations on the population of phosphate-solubilizing microorganisms in the soil ($P > 0.05$) (Table 2), which shows that the chemical characteristics of these fertilizers do not influence the population of these microorganisms. The application of filter cake showed a higher number of these microorganisms, independently if the filter cake was enriched or not, comparing with the treatment without filter cake, which indicates the benefits of the organic compound from filter cake at 45 days after application.

The growth of fungi CFU and total bacteria was favored with the use of Bayóvar reactive phosphate (Table 2). The total fungi population was higher with the enriched filter cake, probably due to the greater amount of these microorganisms from the biofertilizer. The total population of microorganisms was higher with the filter cake, whether or not enriched, comparing to the treatments without its application.

For all microorganisms population, the studied factors (P sources and organic compound) had higher values compared to the control treatment, in which higher populations were obtained in the factorial (Table 2). At 110 days, no significant interactions ($P > 0.05$) between the P sources and organic compound (Table 3) were observed for microbial population. In addition, the effect of P sources on the population of total bacteria and phosphate-solubilizing microorganisms (Table 3) was nonsignificant either ($P > 0.05$). Triple superphosphate and MAP promoted lower numbers of total fungi compared to simple superphosphate and Bayóvar phosphate (Table 3).

Table 2 Total population (CFUg^{-1} dry soil) of bacteria and fungi, phosphate-solubilizing microorganisms (PSM) and total microbial population in soil treated with different phosphorus sources in the presence and absence of filter cake enriched or not with liquid peat and biofertilizers, 45 days after the planting

P sources (P)	Bacteria CFU × 10^6	Fungi CFU × 10^4	PSM –CFU × 10^4	Total CFU × 10^7
Simple superphosphate	6.79	4.39 b	5.75	6.38 b
Triple superphosphate	6.76	4.47 b	5.65	6.43 b
Monoammonium phosphate	6.75	4.26 b	5.70	6.46 b
Bayóvar® phosphate	7.10	4.72 a	5.72	6.66 a
Organic compound (C)				
Filter cake + peat + biofertilizer	7.18 a	4.64 a	5.87 a	6.64 a
Filter cake	7.06 a	4.39 b	5.87 a	6.70 a
Without filter cake	6.32 b	4.36 b	5.38 b	6.11 b
	F Test			
P sources	2.82 ns	8.10**	0.24 ns	5.18**
Organic compound	27.7**	6.98**	15.67 **	52.20**
P x C	0.81 ns	1.43 ns	0.65 ns	0.38 ns
Factorial x control	4.74*	6.73*	8.80**	50.15**
Control means	6.45 b	4.14 b	5.27 b	5.80 b
Factorial means	6.85 a	4.46 a	5.71 a	6.48 a
CV (%)	4.5	4.5	4.3	2.3

Means in the same column followed by different letters differ by the Scott-Knott test ($P < 0.05$). **; *; and *ns* significant ($P < 0.01$); ($P < 0.05$); and nonsignificant, respectively. *CFU* colony-forming units

The lowest values of total microorganisms population were obtained with the MAP. The organic compound promoted an increase in the number of all microorganisms, whether or not enriched, compared to the absence of its application, as well for the factorial means compared to the treatment control (Table 3). Significant interaction was observed for the P content in the soil at 60 (P 0.01) and 110 days ($P < 0.01$) (Table 4). In both periods, the presence of enriched or non-enriched filter cake, the natural phosphate presented the lowest concentrations of soil P in relation to the sources of higher solubility (Table 5). In the absence of filter cake, natural phosphate and MAP were the P sources that showed the lowest levels of soil P compared to simple and triple superphosphates.

It is possible to observe, regarding the organic compound within each P source (Table 5), that MAP was the unique to present effect, and in the absence of filter cake the soil P levels highly decreased when compared with the levels obtained with enriched filter cake and non-enriched filter cake, respectively. At 110 days, the same effect was observed. In addition, at 110 days effect was also observed with the organic compound when triple superphosphate was used, and the highest levels of soil P were obtained with the enriched filter cake.

The other variables had no interaction effect ($P > 0.05$) (Table 4). At 60 days, the dry matter accumulation was higher with triple superphosphate in relation to the other sources, which was not observed at 110 days, in which the P sources had no effect.

P accumulation was higher with simple and triple superphosphate at 60 and at 110 days; the highest value was obtained with MAP and Bayóvar natural phosphate.

The organic compound presented an effect on P and dry matter accumulation in both periods, and the highest values were obtained with the enriched or non-enriched filter cake, in relation to the absence of its application. The treatment control showed lower values for all variables compared to the factorial means (Table 4), which demonstrates the importance of phosphate fertilization and the organic compound on soil P levels, dry matter and P accumulation by sugarcane plants.

Discussion

It can be inferred that the chemical characteristics of each fertilizer can affect the growth of CFU of fungi (Table 3). Increments in the number of soil microorganisms using organic compound (Table 3) was also found in other studies (Ayaga et al. 2006; Babana and Antoun 2006; Gichangi et al. 2009). The number of fungi increased with the use of single superphosphate (Barroti and Nahas 2000); the number of microorganisms was dependent on fertilization, and larger soil populations were obtained with

Table 3 Total population (CFUg^{-1} dry soil) of bacteria and fungi, phosphate-solubilizing microorganisms (PSM) and total microorganisms' population in soil treated with different phosphorus sources in the presence and absence of filter cake enriched or not with liquid peat and biofertilizers, 110 days after the planting

P sources (P)	Bacteria CFU × 10^6	Fungi CFU × 10^4	PSM –CFU × 10^4	Total CFU × 10^7
Simple superphosphate	5.80	4.05 a	5.40	6.47 a
Triple superphosphate	6.02	3.87 b	5.46	6.55 a
Monoammonium phosphate	6.12	3.60 b	5.38	6.21 b
Bayóvar® phosphate	6.22	4.17 a	5.45	6.42 a
Organic compound (C)				
Filter cake + peat + biofertilizer	6.35 a	4.11 a	5.55 a	6.67 a
Filter cake	6.47 a	4.09 a	5.51 a	6.64 a
Without filter cake	5.30 b	3.55 b	5.20 b	5.93 b
	Test F			
P sources	2.03 ns	4.11*	0.36 ns	3.25*
Organic compound	35.40**	9.03**	11.68**	36.76**
P x C	1.68 ns	1.14 ns	0.69 ns	2.01 ns
Factorial x control	5.02*	4.78*	6.67*	5.66*
Control means	5.08 b	3.02 b	4.91 b	5.79 b
Factorial means	6.04 a	3.92 a	5.42 a	6.41 a
CV (%)	6.3	9.4	3.5	3.7

Means in the same column followed by different letters differ by the Scott-Knott test ($P < 0.05$). **; *; and *ns* significant ($P < 0.01$); ($P < 0.05$); and nonsignificant, respectively. *CFU* colony-forming units

mineral or organomineral fertilization, compared to control (not fertilized) (Nuernberg et al. 1984); and the treatments with organic material had increased the number of bacteria and fungi of maize inoculated with phosphate-solubilizing fungus (Nahas et al. 1994).

Therefore, the highest values of CFU observed with the presence of filter cake enriched with peat and biofertilizer and non-enriched filter cake reinforce the hypothesis that the use of organic waste has a fundamental role in the growth and activity of soil microorganisms (Nahas et al. 1994; Jilani et al. 2007; Stark et al. 2007; Jiang et al. 2011; Zhang et al. 2012; Krey et al. 2013), providing better conditions for microbial growth, compared with the isolated application of the P sources.

In our study, the greatest effects on soil microbial population were observed with the organic compound. It is well established that organic matter plays an important role in the maintenance of soil microbial diversity (Jilani et al. 2007; Stark et al. 2007; Zhang et al. 2012), contributing to an increase in biomass of fungi and bacteria (Jiang et al. 2011). This effect may be related to the availability of carbon and nitrogen required for microbial growth and reproduction (Enami et al. 2001); besides, it promotes longer and lasting effects than the chemical fertilizer (Zeng et al. 2007).

Similar results were observed with the application of organic fertilizers in soil cultivated with rice, which resulted in higher populations of total bacteria and fungi and higher microbial activity than in the control or in the treatment with chemical fertilizers (Zhang et al. 2012). Furthermore, the addition of organic waste significantly increases the diversity of functional soil microorganisms (Liu et al. 2009; Pengthamkeerati et al. 2011).

In this study, the application of filter cake increased the population of phosphate-solubilizing microorganisms, indicating that its use as an organic fertilizer in areas cultivated with sugarcane is a viable alternative. In addition, the association of mineral phosphate with microorganisms and organic compound increased the population of phosphate-solubilizing bacteria, actinomycetes populations and P uptake by plants (Shrivastava et al. 2011; Yang et al. 2013). The authors affirm that this method increases the agronomic effectiveness of P sources of low solubility and it is a viable strategy for waste disposal.

Therefore, it is verified the increased availability of P in the soil with the application of high-solubility P sources, especially associated with organic compound (Santos et al. 2009, 2010).

It is noteworthy that at 110 days the soil P levels increased from 9.3 mg dm^{-3}, for the treatment control (Table 4), to 151.3 mg dm^{-3} for the treatment with triple superphosphate enriched with organic compound and biofertilizer (Table 5). This effect is attributed to the use of organic compound in the fertilization, which has helped to

Table 4 Soil phosphorus levels, dry matter (DM) and phosphorus (P) accumulation by sugarcane plants, according to phosphorus sources with or without filter cake enriched with liquid peat and biofertilizers or non-enriched filter cake at 60 and 110 days after planting

P sources (P)	Soil P 60 days	DM	P	Soil P 110 days	DM	P
	mg dm^{-3}	g per plot		mg dm^{-3}	g per plot	
Simple superphosphate	95.4	60.0 b	0.13 a	110.1	102.5	0.24 b
Triple superphosphate	104.1	74.9 a	0.14 a	113.0	109.8	0.23 b
Monoammonium phosphate	78.6	62.1 b	0.10 b	77.6	117.5	0.27 a
Bayóvar® phosphate	21.6	67.2 b	0.09 b	20.3	120.4	0.28 a
Organic compound (C)						
Filter cake + peat + biofertilizer	84.3	70.7 a	0.12 a	101.7	118.5 a	0.28 a
Filter cake	75.5	70.6 a	0.13 a	82.6	122.1 a	0.26 a
Without filter cake	64.9	56.9 b	0.09 b	56.5	97.4 b	0.23 b
	F Test					
P sources	87.8**	4.5*	10.0**	31.5**	1.0 ns	7.5**
Organic compound	8.0**	8.7**	11.4**	11.6**	11.3**	6.4**
P x C	13.2**	1.3 ns	1.6 ns	3.2*	2.1 ns	1.8 ns
Factorial x control	78.5**	52.3**	51.6**	26.3**	75.8**	114.1**
Control means	11.7 b	25.4 b	0.03 b	9.3 b	39.5 b	0.05 b
Factorial means	74.9 a	66.1 a	0.11 a	80.3 a	112.6 a	0.25 a
CV (%)	16.9	14.9	18.9	29.8	13.05	13.3

Means in the same column followed by different letters differ by the Scott-Knott test ($P < 0.05$). **; *; and *ns* significant ($P < 0.01$); ($P < 0.05$); and nonsignificant, respectively

Table 5 Soil P levels at 60 and 110 days comparing the application of P sources with or without filter cake enriched with liquid peat and biofertilizers or non-enriched filter cake

P sources	Organic compound		
	Filter cake + peat + biofertilizer 60 days	Filter cake	Without filter cake
	mg dm^{-3}		
Simple superphosphate	99.0 aA	85.0 aA	102.3 aA
Triple superphosphate	103.7 aA	104.0 aA	104.7 aA
Monoammonium phosphate	112.0 aA	97.7 aA	26.0 bB
Bayóvar® phosphate	22.7 bA	15.3 bA	26.7 bA
	110 days		
Simple superphosphate	120.7 aA	121.3 aA	88.3 aB
Triple superphosphate	151.3 aA	101.7 aB	86.0 aB
Monoammonium phosphate	116.7 aA	90.3 aA	25.7 bB
Bayóvar® phosphate	18.0 bA	17.0 bA	26.0 bA

Means followed by different lowercase letters in columns and uppercase in rows differ by the Scott-Knott test ($P < 0.05$)

reduce the P adsorption (Almeida Júnior et al. 2011; Santos et al. 2010). In addition, the levels of P in the organic compound (1.2%) and in the used dose, 30 t ha^{-1} of filter cake, corresponded to 180 mg dm^{-3} of P. It is important to notice that this P is in the organic form and the process of mineralization must occur to make it available to the plants.

The higher dry matter accumulation and P obtained with superphosphates at 60 days may be explained due to the higher solubility of these sources (Table 4), ensuring greater accumulation of P in the earlier stages of plant

development, and over time, the source with lower solubility presented best results. It was expected that MAP had better results at 60 days, as it is also a source of high solubility; however this fact did occur. The lowest values of total microorganisms population were obtained with the MAP. The high solubility of these P sources, together with the ammonia contained in the MAP (NH_4H_2PO), may have caused changes in the soil chemical properties, interfering in the population of these microorganisms. The highest values of P accumulation with natural phosphate at 110 days are not in accordance with the soil P levels, since the natural phosphate showed lower levels of available P in relation to the other sources, although it can be inferred that these levels were sufficient for plant growth and P accumulation, and it did not lead to lower dry matter production.

More pronounced effects of fertilizers on soil properties than in dry matter accumulation and P uptake by plants were observed (Krey et al. 2013). In alkaline soils was an increase in soil P levels and P uptake by plants, resulting in higher growth and productivity of wheat and maize, compared with the control (soil) and with isolated application of phosphate rock or *Penicillium oxalicum* (Singh and Reddy 2011). The authors verified that the combination of phosphate rock with *Penicillium oxalicum* can replace the chemical fertilizer in alkaline soils. This result is associated with the capacity to release organic acids and pH decrease (Relwani et al. 2008), especially in alkaline soils.

In this study, there were no significant enhancements in the evaluated attributes with the addition of biofertilizer to the organic compound (Tables 2, 3 and 4). A fact that may explain this lack of response is that microorganisms play an important role in P solubilization in alkaline soils (mainly P bound to Ca), when compared with acid soils (predominance of P bound to Fe and Al). Another reason would be that only the filter cake was sufficient to raise the soil microbial population, which presented 2.97×10^5 CFUg^{-1} of total bacteria, 2.5×10^2 CFU g^{-1} of P-solubilizing bacteria and 2.5×10^3 CFU g^{-1} of total fungi. The benefits of organic compound associated with the mineral P fertilization on soil P levels and P uptake by plants have also been reported (Mahfouz and Sharaf-Eldin 2007; Takeda et al. 2009).

Several studies with different organic compounds and biofertilizers associated with phosphate fertilizer showed the benefits on the availability of P levels in soil and productivity of different crops, such as mung bean (*Vigna radiate*) (Biswas and Narayanasamy 2006); maize (Abou-el-Seoud and Abdel-Megeed 2012; Krey et al. 2013); sugarcane (Shankaraiah et al. 2000; Stamford et al. 2006); banana (Ezz et al. 2011); and fennel (*Foeniculum vulgare* Mill.) (Mahfouz and Sharaf-Eldin 2007).

Therefore, the addition of organic compounds leads to increased formation of organic forms of P, whereas addition of P promotes the formation of inorganic non-labile P (Malik et al. 2012). The organic P compounds of microbial origin are readily hydrolysable and may become available to the plants (Kouno et al. 2002). Thus, microbial assimilation of P from less labile forms can be effective in improving the efficiency of P use by crops on soils with high rates of P fixation, even though P assimilation by microorganisms contributes to the maintenance of lower soil P levels (Ayaga et al. 2006).

Most of the solubilized P is temporarily immobilized by the microorganisms in the microbial cells. The microbial biomass P (organic P) acts as a slow fertilizer releaser because of its slow decomposition rate and provides available P to the plants for a longer period, reducing the rates of adsorption and/or precipitation of soil minerals, such as in the case of highly soluble phosphates (Biswas and Narayanasamy 2006).

Conclusion

The organic compound increased the population of soil microorganisms, the soil P levels, the P and dry matter accumulation by plants, regardless of the source of P, although the enrichment with biofertilizer and liquid peat did not enhance this effect.

Compliance with Ethical Standards

Conflict of interest There is no conflict of interest.

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